

Examiner reports

Q2.

Not all candidates appeared to have time to attempt this question and often solutions appeared rushed rather than thought out. However many of those who did attempt the question solved the differential equation in part (a) correctly. Most candidates separated the variables correctly using conventional notation, but then there were often errors in the integration. Common ones included the omission of a constant, incorrect integration of both x and $\cos 2t$ or use of degrees instead of radians when evaluating the constant. Some candidates just substituted the given conditions straight into the differential equation, producing nonsense.

In part (b)(i) some candidates substituted $\pi/4$ into dx/dt instead of their solution from part (a). Another error was to substitute $13\pi/4$ for t . There were, however, some correct solutions showing understanding and most candidates who attempted this did show they intended to work with $\sin 26$, even if they were trying to use it in an incorrect expression or with their calculator in degrees. Most gave answers to the accuracy required by the question.

In part (b)(ii) those who had successfully solved the differential equation in part (a) usually earned the method mark here. However correct answers were rare, as many candidates arrived correctly at a negative value for $2t$, and just ignored the negative sign, rather than considering the next positive solution. Another common error lay in using the wrong calculator mode once more.

Q3.

There were very mixed responses to this question. Some students worked through this very easily, but others found it difficult to get started and produced very little meaningful work. A very common issue was mixing up aspects of the horizontal and vertical motion. For example equations like $38.4 = V \sin \alpha t - 4.9t^2$ were seen. Another incorrect approach was to assume that the ball was at its maximum height half way through the motion.

Some students attempted the question in an order that did not match the parts of the question. Typical of this was simply stating $V \cos \alpha$ in part (a) and then finding the value of 16 while doing part (b). It should be noted that if an answer of this type was required it would have been asked for in terms of V and α . (However please note that full marks were given to these students). Many also found the angle 39.1° while working on part (b). As a general principle, students should make sure that they give the answer that they intend for each part where they have written the part letter (eg (a)) on their script. In this question it did not cause issues for the marking, but in another case it could be more problematic.

Q4.

This question was also answered well by virtually all students who showed that they were familiar with the techniques involved in answering this question. In part (b)(ii) students sometimes failed to find that the value of $-16\pi \sin \pi \mathbf{i}$ was zero; this was usually due to students using the angle in degrees rather than radians. The magnitude of $\mathbf{F}$ was often given as -216 rather than $+216$. In part (c) the majority of students integrated to find the position vector but a common error was to use the vector $4\mathbf{i} - 2\mathbf{j}$ as the $+c$ term. A considerable number of students found the $+c$ term correctly but simplified their final $\mathbf{j}$

term to $-(3t^3 + 79)\mathbf{j}$ rather than $+(79 - 3t^3)\mathbf{j}$.

Q5.

The students approached this question either using the cosine and sine rules or by finding components. The big issue for those who used the cosine rule was that some students used 30° and did not realise that in the triangle they should be considering that it was a 150° angle that was required. This approach made it very difficult for them to gain marks on the question, whereas those who had the correct angle often went on to gain full marks.

Those who used an approach based on components could do well, but there were a couple of common mistakes, typically due to getting mixed up with the values that they had obtained. A common error of this type was to use 70 instead of 104.6 in part (b) when finding the angle. All the students who had difficulties with this question would probably have been helped by better diagrams.

Q6.

Students tended to do either very well or very badly on this question. This projectiles question, which did not contain a specified angle, seemed to confuse many students and some even tried to find an angle that they could use in some way. The problems were generally focussed on a confusion between the horizontal and vertical motion and the inability of students to relate the information given to these directions. For example in part (a), some students used 20 m to try to calculate the time, or in some cases the length of a line from the top of the cliff to the boat. In part (b), which required consideration of the horizontal motion an acceleration of 9.8 m s^{-2} was sometimes included. Similarly in part (c), the same type of confusion was present.

Also in Part (c), there were a few students who correctly found the two components of the velocity, but who did not then find the speed of the car.

Q7.

Part (a) was often done very well by students, with a reasonable number of students gaining all 11 marks. There were a few issues with the force diagram, with missing or additional forces. Many students seemed to have been helped by the answer that was given for the normal reaction and this in turn enabled many to find the friction correctly. Finding the acceleration was also done well, although there were a few sign errors and some students simply used 150 instead of $150 \cos 20^\circ$. Part (b) was much more demanding and there were far fewer correct responses. The most common error was not to take account of the changed angle when finding the normal reaction and to simply continue to use the value of 243 from part (a).

In part (c) there were some good responses, but a fair number of students incorrectly stated that the tension would be greater for a greater constant speed.

Q8.

- (a) Almost all students were familiar with the equations of motion of a projectile and they were able to use them to find the equation of the projectile's trajectory in terms of x , u , g and $\tan\theta$.
- (b) (i) Most students interpreted the context of the question correctly in relation to the given co-ordinates axes and they were able to make the appropriate substitutions into their equation of the trajectory. They were then able to solve the quadratic equation in $\tan\theta$. However, many students lost the last accuracy

mark because they stated the given values of g without writing down the unrounded values to at least four significant figures.

- (ii) This part proved more challenging for some students. Those who answered this part correctly were able to find the horizontal and vertical components of the velocity of the ball at the instant of reaching the basket and then used trigonometry to find the required angle. Students needed to specify the direction of the motion of the ball by giving a correct acute angle and stating whether the angle was made with the horizontal or the vertical.

Q9.

There were very many good attempts at both parts of part (a) with a lot of correct answers. However, there was quite a bit of confusion with part (b). A very common wrong answer was 686 N, and it appears that there was some confusion between the words 'resultant' and 'reaction'. There were some good solutions to part (c), although this part of the question was more demanding. A considerable range of incorrect responses were seen, including the omission of g , as in $70 \sin \alpha = 58.8$, the use of $\cos$ instead of $\sin$ or the use of the normal reaction from part (b), as in $686 = 70g \sin \alpha$.

In the second part of part (c), it was not unusual to see the 30 N force with the wrong sign which produced the following incorrect equation: $70g \sin \alpha = 58.8 - 30$. In part (d), there were many good answers where the candidates linked the resistance to the speed of the cyclist. However, some simply said things such as "the resistance will not be constant" and did not give a reason to support this answer.

Q10.

This question proved quite difficult, but candidates did tend to pick up marks as they progressed through their attempts. Many candidates gained 1 or 2 marks in part (a) for their attempt to find the angle. Some, however, tried to use the speed of 8 rather than the distances and scored zero.

In part (b), a reasonable number of candidates produced a quadratic equation, but often with incorrect signs, $8t$ instead of $8\sin \alpha t$ or the wrong distance. Once an error of this type was made, they began to lose marks, although those who showed how they solved their quadratic equation often gained the method mark. Even those with poor attempts at part (b) were able to gain some marks in part (c) by using $8\cos \alpha t$ and subtracting their answer from 10, although there were very few completely correct solutions.

Q11.

This question was done very well by candidates and there were many full correct solutions. The main source of errors in part (a) was due to using incorrect combinations of $\sin$ or $\cos$ and the angle that the candidates decided to work with. In part (b) some candidates used trigonometry while other used Pythagoras' Theorem. In the final part some candidates multiplied by 9.8 rather than dividing by it.

Q12.

The candidates made very little progress with this question and did not seem to be able to deal with the resolving required.

Q13.

There were many correct answers to part (a) and the candidates were clearly helped by the presence of the printed answer. In most cases the arguments given were sound, but in others this was not the case. For example some candidates simply wrote statements such as $22.4 \sin \theta - 19.6$ and never included an equals sign or a zero. This was regarded as an important omission.

There were also lots of good answers to part (b), but some basic errors, such as the omission of a negative sign with g or the use of $\sin(0.875)$. In part (c) there were also some good answers. Some candidates did not realise that the time of flight could be found by doubling the 2 seconds that was given, and did quite a lot of work finding the time of flight. In some cases there were errors in the calculation of the distance. For example an

incorrect application of $s = \frac{1}{2}(u + v)t$ was seen with statements such as $s = \frac{1}{2}(10.8 + 0) \times 4$.

The approaches to part (d) were many and varied. There were quite a number of good responses, but often candidates did not really get very far with this part of the question. The most common error was for candidates to use a distance, for example $19.6 - 5 = 14.6$, with the initial speed of $22.4 \sin \theta$.

Q14.

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Q15.

The force diagram asked for in part (a) was generally well drawn, but some candidates did lose marks for not including arrow heads or appropriate labels. The candidates who understood that they should resolve vertically to find the reaction force did well although some ended up with $R = 78.4 + 0.5T$ or equivalent. Others produced very confused responses.

Those who had a correct expression for the normal reaction often made some progress in part (c). There were quite a number of candidates who made promising starts but then made algebraic errors when solving for T . There were a range of errors that were made in setting up an equation of motion. These included only using the friction force, using T instead of $T \cos 30^\circ$ and omitting the ma term.

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The approaches to part (d) were many and varied. There were quite a number of good responses, but often candidates did not really get very far with this part of the question. The most common error was to see candidates use a distance for example $19.6 - 5 = 14.6$, with the initial speed of $22.4 \sin \theta$.

In the final part, there was a tendency for the candidates to find the resultant of two velocity components, as they did not realise that for the minimum speed the vertical component would be zero. In some case the correct final answer was obtained from expressions such as $\sqrt{10.8^2 + 0^2}$.

Q17.

The great majority of the candidates were able to answer both parts of this question correctly. However, a small number of candidates lost the last accuracy mark which was available for part (b) by not showing enough working. Centres should advise their candidates to show sufficient evidence of working when the answer to a question is given on the question paper.

Q18.

Part (a) was done successfully by the vast majority of the candidates. The force diagrams in part (b) were usually good, but the most common errors were to omit either the normal reaction or friction forces. Also some candidates used lines rather than arrows to represent the force.

In part (c), the candidates were often able to produce the printed answer, but in some cases did not show enough working to justify the award of full marks. For questions like this, candidates should be advised to show an answer to more than three significant figures before giving their final answer.

Those who started part (d) by resolving to find the friction force generally did well and completed this part of the question. A number of candidates did make significant errors by assuming, for example, that the friction force was $2g$ or in some cases $4g$.

Q19.

A large proportion of the candidates produced a quadratic equation, but there were several potential errors caused by setting up the equation in the wrong way. A few used the 1 or the 1.5 but not both, some used one with the incorrect sign and ended up with a 2.5 term instead of a 0.5 term, and some used 0.5 but with the wrong sign. Other methods that were seen included finding the time to move up and the time to move down and then adding them together. Some candidates lost one mark due to a lack of intermediate working and a failure to show how 1.30 was obtained from their quadratic equation. Again, showing an answer to more than three significant figures before quoting the final answer would have helped them.

Part (b) was usually done well with candidates using the printed answer from part (a).

In part (c), some candidates used the required approach to find the speed of the arrow, but many did not realise how to approach this part of the question. A few candidates made sign errors so that they obtained a vertical component of velocity that was too great

($12\sin 30^\circ + 9.8 \times 1.3 = 18.74$). A few candidates only considered one component of the velocity. Those who had done well on part (c) were usually able to obtain the marks for part (d) too.

There were a lot of good answer to part (e), and in some cases this was the only place on the question where the candidate gained any marks.

Q20.

Some candidates produced very good clear solutions to this question. While there were many good force diagrams, some were confused and did not make use of arrows to indicate the forces. There were some issues with the direction of the tension and the 500 N force.

In part (b), the candidates who could start with a clear statement of the equation of motion, usually completed the question and gained full marks. For those who realised that an equation of motion was required, the most common errors were to omit a force or to take the wrong component. A few candidates omitted the ma term and treated the problem as if the forces were in equilibrium.

Q21.

This question was more demanding and candidates sometimes confused the horizontal and vertical aspects of the motion, for example by including the 250 horizontal component of velocity in an equation for the vertical motion.

However, part (a) was generally done well. In part (b), the confusion between the horizontal and vertical components of the motion was most apparent,

with expressions like $s = 250 \times 4 + \frac{1}{2} \times 9.8 \times 4^2$ being seen. Candidates who took this approach in part (b) then often made little progress in the later parts of the question. Similar problems were evident in part (c) when candidates tried to calculate the vertical component of the velocity. In part (d), some candidates used distances rather than components of the velocity to calculate the angle. In some cases there was a mixture of a velocity and a distance!

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Q22.

While there were a number of good solutions to this question, there were also a very significant number of weak attempts. In part (a), many candidates could write down either a correct horizontal or vertical equation, but then were unable to derive the required answer correctly. Some could get one equation but not the other. In some cases the equations were confused, for example containing both a and g in the same equation. In part (b), there were again a number of correct solutions. The most common error here was to work on the assumption that P would also be zero. The candidates who answered correctly produced a variety of solutions, which included finding an expression for Q and starting again as if there was only one string.

Q23.

This question was answered very well by the great majority of the candidates. Part (b)(i) of the question was prone to errors for some candidates who did not take sufficient care in getting a correct simplified quadratic equation in $\tan \theta$ and solving the equation to find the

two possible values of θ . Some of these candidates wrote 89.8° for a possible value of θ and did not seem to question or check the reasonableness of their answer. For part (b)(ii), almost all candidates chose their smaller angle for the answer, but many could not provide a convincing reason for their choice.

Q24.

The candidates seemed to be helped by the printed answer in part (a), but some obtained the printed result from the use of equations such as $5 = \frac{1}{2}(-9.8)t^2$. There were also some candidates who did not justify all of their working. There were a number of candidates who found it difficult to work with the two components of the velocity in parts (b), (c) and (d).

A reasonably common error was to use 5 instead of 15 in part (b) and then 15 instead of 5 in part (c). Some candidates also stopped after obtaining the vertical component in part (c) and did not go on to find the speed. Those who did find the components correctly were usually able to go on to find the angle correctly.

Q25.

This question was also done well by the majority of candidates. While most candidates were able to produce the printed answer for part (b), there were some issues with the force diagram and finding the tension in part (c). The two main problems with the force diagram were including extra forces, for example friction or simply poor diagrams without arrows or labels. In part (c) there were some confused responses and some candidates tried to resolve the normal reaction from part (b).

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Q27.

In part (a) many candidates lost marks for not including arrow heads or due to poor labelling of the forces. Several candidates did not show the two tensions as being equal on their diagrams. There were quite a few good attempts at part (b), with attempts at resolving vertically. Part (c) proved to be very challenging, although there were a number of good solutions. A number of candidates gained an M1 mark for the use of the friction law. The most common errors were to omit one force in their equation or to not include the ' ma ' term in their equation of motion.

Q28.

This question was found to be quite difficult by many of the candidates. In part (a) there were a number of incorrect force diagrams, including errors such as including a force acting directly down the slope. In parts (b) and (c) there were a lot of confused answers, mainly due to incorrect attempts at resolving. For example, in part (b), one error was to resolve the normal reaction into components instead of the weight. However, many candidates gained full marks for part (d), where follow-through marks were awarded when the candidates' values from parts (b) and (c) were used correctly in the friction law to find a coefficient of friction.

Q29.

Very many candidates were able to prove the result in part (a). Most considered the vertical component of the velocity and found the result easily. Some tried to find the time of flight first. In some cases these candidates halved their answer, but some did not do this or did not make it clear. A very small number of candidates used equations that they had learned for the time of flight or similar, but did not justify these results and so did not gain the marks. Candidates should not use learned formulae for the range etc in questions like this.

There were many different approaches to part (b)(i). Quite often candidates could produce a correct equation as a starting point, but failed to simplify it correctly. Often the candidates ended up with a mixture of $\sin \alpha$ and $\sin^2 \alpha$ terms. There were many minor arithmetic or algebraic errors. In part (b)(ii), many candidates used the incorrect time: their

approach was based on a time of $\frac{3 \sin \alpha}{2}$ instead of $3 \sin \alpha$.

Q30.

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approach was based on a time of $\frac{3 \sin \alpha}{2}$ instead of $3 \sin \alpha$.

Part (c) was often done well, but quite a few candidates incorrectly talked about the projectile having no mass.

Q31.

This question was answered very well and proved to be an easy source of marks for many candidates.

In part (a)(i), the candidates seemed comfortable with the kinematic equations of motion

and the identity $\frac{1}{\cos^2 \theta} = 1 + \tan^2 \theta$. Very few candidates substituted 20 instead of –20 for y in answering part (a)(ii).

Some candidate lost an accuracy mark in part (b)(i) for premature rounding of the solutions of the equation for $\tan \theta$. Other candidates who wrote down inaccurate values for $\tan \theta$ but used the accurate values to find the values of θ did not lose any accuracy mark. Centres should remind their candidates to give the **final** answer to questions to three significant figures, unless stated otherwise, as stated in the instructions on the front page of the question paper. In part (b)(ii), a small number of candidates gave the result to less than three significant figures.

Q32.

This question also provided more challenge for the candidates. In this question a number of candidates did assume that the acceleration due to gravity was positive.

In part (a), quite a number of candidates found the time of flight and then used half of this value to calculate the maximum height. There were however a number of cases where the candidates did not half their value for the time of flight and so produced solutions which did not relate to the maximum height.

Part (b) was much more demanding and relatively few correct solutions were seen. Those candidates who identified the required method were able to do well, but the majority made no attempt or produced working which could not lead to the required answer. Similarly part (c) was found to quite difficult, but some candidates did use the answer from part (b) to put together correct solutions for part (c) when they had been unable to do part (b).

Q33.

There were many correct force diagrams on the scripts, but some candidates lost marks due to missing arrows or poor labelling. The printed answer in part (b) seemed to help the candidates as there were many correct answers. It was clear that in some cases candidates had needed to change their approach to obtain the correct value. There were also many good answers to parts (c) and (d), but some candidates included extra forces which meant that their friction value was wrong. A few candidates seemed to confuse the coefficient of friction and the friction force.

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In part (a), quite a number of candidates found the time of flight and then used half of this value to calculate the maximum height. There were however a number of cases where the candidates did not half their value for the time of flight and so produced solutions which did not relate to the maximum height.

In part (b), as might be expected some candidates had difficulty with the signs in their equations. Those who were able to form a reasonable quadratic often went on to solve this to find a value for the time. However quite a few candidates did not show both solutions of the quadratic. It is important for candidates to show both solutions and

indicate which one they have selected. Some candidates did not show enough intermediate working to gain full marks in this “show that” type question.

A few candidates used a method which involved finding the time taken to move from the highest point to the point where it was caught and adding this to the time to reach the maximum height. This method was often used by candidates who had found a time in part (a).

In part (c) a large number of candidates simply found the vertical component of the velocity when the ball was caught and did not consider the horizontal component at all.

Q35.

Parts (a) and (b) were found to be fairly straightforward by candidates, but part (c) was much more demanding. The candidates often calculated the vertical component of the velocity in part (c), but were unable to use this correctly in further calculations. There was often confusion about how to calculate the speed and the direction of motion.



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